Quantum Traffic Gridlock

Scenario-independent QUBO and QAOA signal scheduling for two linked intersections

72.0
Exact optimum C* on scenario R, enumerating all 256 schedules

110.0
Raw QAOA mean official cost over 20,480 shots (random mean 151.25)

72.0
Hybrid: best official cost among sampled bitstrings (labelled separately)

0.520
Mean raw-QAOA lift on R (min 0.515); 8 qubits, p = 1, depth 16

1 Traffic model and notation

Two intersections (A and B) are scheduled over four slots. Let x[A,t] and x[B,t] be binary: **1** selects north-south (NS) and **0** selects east-west (EW). The canonical eight-bit order is `A1,A2,A3,A4,B1,B2,B3,B4`. The official scorer adds arrivals and prior-slot transfers, serves up to capacity C on the green approach, and scores the remaining queues. Its transfer is $\mathrm{floor}(f \cdot \mathrm{departures} + 0.5)$, delivered to B at the start of the next slot; transfers after slot 4 leave the network.

$$\text{Official cost} = \text{delay} + w_{\text{ovf}} \cdot \text{overflow} + w_{\text{sw}} \cdot \text{switches}$$

2 Scenario-independent QUBO

For each approach-slot, the builder forms a linear queue proxy $q_i(x) = c_i + a_i^{\mathrm{T}}x$. It accumulates external arrivals, approximates green service as $C{\cdot}x$ for NS or $C{\cdot}(1 - x)$ for EW, and models A-to-B arrivals from the next slot onward as $f{\cdot}C$ times the linked-phase green indicator. Unlike the official model, these proxy terms do not clip service to available queue and do not round fractional transfers. **The scorer remains authoritative for schedule costs.**

$$C(x) = \Sigma_i q_i(x) + \gamma\, \Sigma_i (q_i(x) - \alpha S)^2 + w_{\text{sw}}\, \Sigma_t (x_t \oplus x_{t+1}), \qquad \alpha = 0.50$$

Delay has weight 1; switching uses the scenario's w_{sw} and the exact binary identity $x \oplus y = x + y - 2xy$. The quadratic congestion term approximates the scorer's max(0, queue − S) overflow hinge. For $q = c + a^{\mathrm{T}}x$, expansion uses $x_i^2 = x_i$: the constant is c_{shift}^2, diagonal terms are $(2c_{\text{shift}}a_i + a_i^2)$, and pair terms are $2a_ia_j$, all multiplied by γ. No auxiliary variables are used; there are **eight QUBO bits**.

The penalty scale is $\boldsymbol{\gamma = 2{\cdot}w_{\text{ovf}} / S^2}$. The multiplier 2 was retained from the first 150 scenarios of a 200-scenario random bank (seed 20261015; arrivals 0-8, capacity 4-6, storage 5-8, overflow weight 2-6, switch weight 0.25-1.5, link fraction 0.4-0.8). The center α was then compared at 0.25, 0.50 and 1.00 on a separate 100-scenario bank (seed 77441931); **α = 0.50** won by mean regret, P90, worst regret, then correlation. R and H were excluded from both tuning banks. The seed-77441931 set was used for candidate selection and is described as a holdout in the source report.

Scenario bank	Seed	Size	Used for
Random bank (arrivals, capacity, storage, weights, link fraction)	20261015	200 (first 150 used)	γ multiplier
Candidate-selection bank	77441931	100	α ∈ {0.25, 0.50, 1.00}
Depth-ablation bank (excl. R/H)	20261016	16	p = 1 vs p = 2
Hidden-range bank for full ablation	20261017	4 + R	900-configuration ablation

Limit of the surrogate. Linear service can make proxy queues negative, fractional transfers differ from half-up integers, and a smooth square is not the scorer's clipped overflow hinge.

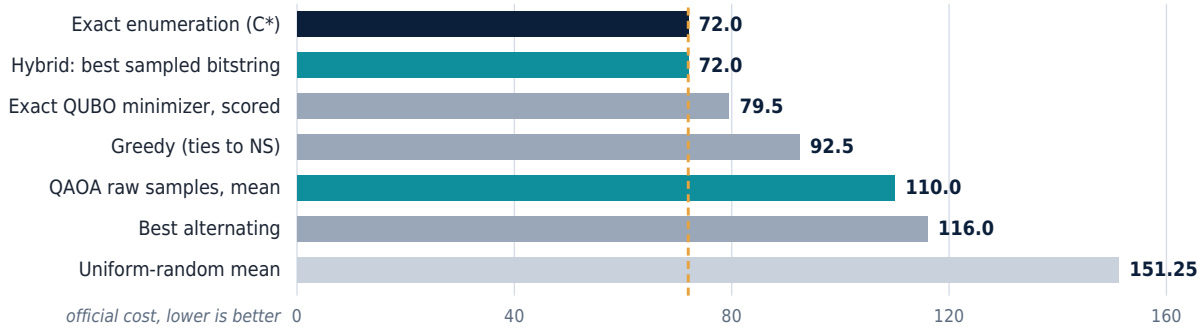
3 QAOA design and R benchmark

With $z_i = 1 - 2x_i$, the upper-triangular QUBO becomes an Ising energy. Exhaustive parity over the 256 assignments has maximum error **5.68e-14** for R. Each layer starts with Hadamards, applies RZ/RZZ cost rotations and RX mixers.

$$E(z) = K + \Sigma_i h_i z_i + \Sigma_{i<j} J_{ij} z_i z_j$$
$$K = \text{offset} + \tfrac{1}{2}\,\Sigma_i Q_{ii} + \tfrac{1}{4}\,\Sigma_{i<j} Q_{ij}, \quad h_i = -Q_{ii}/2 - \tfrac{1}{4}\,\Sigma_{j\neq i} Q_{ij}, \quad J_{ij} = Q_{ij}/4$$

Production uses **p = 1**, **COBYLA**, a seeded linear-ramp initialization with uniform jitter ±0.01, and a hard 300-objective-evaluation cap. The Ising cost coefficients are divided by their uniform-sampling standard deviation before constructing the cost rotations; the QUBO and official scores are unchanged. The depth ablation used 16 random scenarios (seed 20261016, excluding R/H): normalized gaps were 0.09161 for p = 1 and 0.07907 for p = 2, with mean optimizer times 0.1596 s and 0.3076 s. We fixed p = 1 to reduce layer and runtime cost. Final sampling uses exactly **4,096 shots** and the run seed. Qiskit prints `c7..c0`; decoding reverses that string to `A1..A4,B1..B4`.

R method comparison (C* = 72.000; C_{rand} = 151.250)



Method	Schedule / interpretation	Official cost
Best alternating	10101010	116.000000
Greedy (ties to NS)	01111001	92.500000
Exact enumeration	11101010	72.000000
Uniform-random mean	-	151.250000
Exact QUBO minimizer (officially scored)	11101001	79.500000
QAOA raw samples (mean E_Q, 5 seeds)	5 x 4,096 raw shots; E_Q mean	110.006592
Hybrid (best official cost among all sampled bitstrings)	11101010	72.000000

The raw QAOA row is the official-cost mean over all 20,480 raw shots: **110.006591797**. Seed 1's most-frequent raw schedule is `11100110` (cost 99.5). The separately labelled hybrid row selects the best official-cost schedule present in sampled bitstrings across the five runs; it found `11101010` at 72.0. **This postselection is not the raw-QAOA mean.**

Per-seed raw-QAOA metrics

Seed	E_Q	Lift	Elite enrichment	Optimizer evals
1	110.409912109	0.515332339	3.384615385	63
7	110.024658203	0.520193587	3.360576923	74
21	110.095947266	0.519294041	3.216346154	62
42	109.600585938	0.525544657	3.485576923	58
101	109.901855469	0.521743149	3.350961538	55

Minimum lift clipped to [0, 1] over the five seeds: **0.515332339314**. Lift = (C_{rand} − E_Q) / (C_{rand} − C*). Elite enrichment is the fraction of raw shots in the 13 lowest-cost schedules (ties by ascending bitstring), divided by 13/256.

Scenario H: locally proposed stand-in

Scenario	Qubits / depth	Most-frequent raw sample (cost)	Exact optimum	Hybrid sample
R (official)	8 / 16	11100110 (99.5)	72.0	72.0
H (local proposal)	8 / 16	11110000 (178.0)	163.0	163.0

No official H was available to us. We generated `scenario_H.json` from the hidden-range parameters to exercise the pipeline; **it is a stand-in, not the official H**, and its numbers should not be read as official results. H used 46 of 300 optimizer evaluations and 0.227 s of QAOA runtime (seed 1, 4,096 shots, as for the R main run). Both runs write `qubo.json`, `circuit.qasm`, `counts.json` and `schedule.json`; repeating a run with the same seed reproduced the counts and schedule exactly.

4 Ablations, generalization, and limitations

The full ablation covers **p = 1..4**, gamma multipliers **×0.5 / ×1 / ×2**, and **linear-ramp / random / fixed** initialization, crossed over five seeds on R plus four fresh hidden-range scenarios (bank seed 20261017). There were **900 configurations**; each had an 80-evaluation cap. Values below average all scenarios, seeds, and settings for the other factors. Ideal-statevector expectations are used for this ablation; the separate generalization table uses the normal 4,096-shot CLI.

Factor	Level	Expected official cost	Lift	Elite enrichment
Depth	p = 1 (production)	164.161	0.32547	2.5152
	p = 2	162.561	0.35088	2.6015
	p = 3	162.417	0.35202	2.6020
	p = 4	165.885	0.28003	1.9674
Gamma	×0.5	163.238	0.32871	2.5201
	×1	164.178	0.32223	2.3729
	×2	163.852	0.33036	2.3716
Init	linear-ramp (production)	160.697	0.39216	2.5977
	random	170.562	0.18812	1.8795
	fixed	160.009	0.40102	2.7874

Shaded cells mark the lowest expected official cost within each factor.

Unmodified default pipeline on R and the four random scenarios, five seeds each

Scenario	Mean E_Q	Mean lift	Min clipped lift	Mean enrichment	Max evals	Max QAOA s
R	110.006592	0.520422	0.515332	3.3596	74	0.279
G001	357.872267	0.443755	0.435543	3.0856	77	0.281
G002	170.316777	0.393655	0.388075	2.7529	77	0.330
G003	76.981818	0.524564	0.521080	3.1913	73	0.272
G004	77.884200	0.326826	0.320738	2.9279	96	0.322

Honest limitations.

• On R, the exact QUBO minimizer is `11101001` with official cost 79.5, **10.42% regret** relative to C* = 72. Its overlap with the official elite set is 4/13.
• In the selected-center 100-scenario bank, the worst exact-QUBO regret is **47.11%**: a heavy tail despite the lower mean.
• Raw QAOA R mean lift is 0.520422 and its minimum clipped lift is 0.515332. The hybrid happened to find the exact optimum in these samples, but raw sampling is evaluated separately from the raw mean.
• CVaR was not implemented. No official scenario H was available; the local H above is a stand-in. Model-mismatch sources include unclipped linear service, fractional unrounded transfers, and the smooth overflow penalty.

Source table details: candidate α = S/2 R regret 10.42 and R elite overlap 4/13; worst holdout regret 47.11. Generalization seed totals and all per-seed rows are in `outputs/block7/generalization.json`. Reproduce: `python run.py --scenario scenario_R.json --seed 1 --out outputs/R` and `python run.py --scenario scenario_H.json --seed 1 --out outputs/H`.